This is your fourth homework assignment. I strongly encourage you to typeset it using the LaTeX template provided. It is due by Wednesday, February 26th

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The following questions are on the Simplex Method.

(1) Consider the following linear programming problem in standard form:

$$\begin{aligned} \text{minimize } & 2x_1 - x_2 - x_3 \\ \text{subject to: } & 3x_1 + x_2 + x_4 = 4 \\ & 6x_1 + 2x_2 + x_3 + x_4 = 5 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

Consider the basis $B = \{3, 4\}$.

(a) Compute the basic feasible solution, $\mathbf{x}_B$, corresponding to this basis.

(b) Is this solution optimal? Justify your answer.

(2) Use the simplex method to solve the following problem:

$$\begin{aligned} \text{minimize } & -x_1 - x_2 - 3x_3 \\ \text{subject to: } & x_1 + x_3 = 1 \\ & x_2 + x_3 = 2 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

starting with the basis $B = \{1, 2\}$. For each iteration of the method, clearly display B, $\mathbf{x}_B$, all δ_j used, $\epsilon_{\max}$ and i^*. *(See the lecture notes for definitions of these quantities.)*

The following questions are on Convex Optimization, Gradient Descent and Newton's Method.

(3) Prove that if $g(\mathbf{x}) : \mathbb{R}^d \to \mathbb{R}$ is convex and $f(x) : \mathbb{R} \to \mathbb{R}$ is convex and non-decreasing then $f(g(\mathbf{x}))$ is convex.

(4) Recall that when analyzing gradient descent for a function $f(\mathbf{x})$ we introduced the following linear approximation with proximal term:

$$L(\mathbf{x}) = f(\mathbf{x}_k) + (\mathbf{x} - \mathbf{x}_k)^\top \nabla f(\mathbf{x}_k) + \frac{1}{2\alpha_k}\|\mathbf{x} - \mathbf{x}_k\|_2^2$$

show that $\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k)$ is the solution to $\text{argmin}_{\mathbf{x} \in \mathbb{R}^d} L(\mathbf{x})$.

(5) **Optional** Recall the least squares problem $\mathbf{x}_* = \text{argmin}\|A\mathbf{x} - \mathbf{b}\|_2^2$ and let $f(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|_2^2$, so that the problem becomes $\mathbf{x}_* = \text{argmin} f(\mathbf{x})$. At the beginning of the quarter we established two methods for solving this problem: using the normal equations and using the QR decomposition. We also established that $\nabla f(\mathbf{x}) = 2A^\top (A\mathbf{x} - \mathbf{b})$, so one can also use gradient descent to solve this problem. Write code for solving the least squares problem using these three methods, for increasing size of matrix A. Which method is fastest for very large A?

(6) In this question we will motivate part of the convergence theorem for gradient descent. We'll proceed by stages, but you only need to provide an answer for parts (b) and (c).

(a) If a function $f(x)$ is L Lipschitz differentiable then for any $\mathbf{x}, \mathbf{y}$ in its domain:

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{L}{2}\|\mathbf{y} - \mathbf{x}\|_2^2 \tag{1}$$

(You do not need to check or prove this.)

(b) Let $\mathbf{x} = \mathbf{x}_k$ and $\mathbf{y} = \mathbf{x}_{k+1}$ in (1). Use the fact that for Gradient Descent $\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k)$ to simplify as much as possible.

(c) Conclude by explaining why $\alpha < 2/L$ guarantees descent (*i.e.* guarantees that $f(\mathbf{x}_{k+1}) < f(\mathbf{x}_k)$.

(7) In class we observed that Newton's Method (with step size 1) finds the minimizer of a quadratic function $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top Q\mathbf{x} + \mathbf{x}^\top \mathbf{b}$ in a single step. We shall assume that Q is positive definite. Here we will prove this.

(a) Compute $\nabla f(\mathbf{x})$ and $\nabla^2 f(\mathbf{x})$, then write down and simplify the Newton update:

$$\mathbf{x}_1 = \mathbf{x}_0 - \left(\nabla^2 f(\mathbf{x}_0)\right)^{-1} \nabla f(\mathbf{x}_0).$$

(b) Use the condition that for $\mathbf{x}_*$ to be a local minimizer, $\nabla f(\mathbf{x}_*) = 0$, to solve for the unique local minimizer of $f(\mathbf{x})$. *(Because $f(\mathbf{x})$ is strongly convex, this is also the unique global minimizer).*

(c) Now verify that, no matter what $\mathbf{x}_0$ is, $\mathbf{x}_1$ is equal to $\mathbf{x}_*$.